

Phase 1.3 Summary

Phase 1.3 focused on implementing the tilemap system and integrating crawler interactions with multiple tile layers. The key tasks and features completed are as follows:

Tilemap & Layer System

- Introduced a **multi-layer tilemap** architecture.
- Defined four layers:
 - `LAYER_BASE` for walkable floor tiles.
 - `LAYER_WALLS` for solid and digestible obstacles.
 - `LAYER_HAZARDS` for environmental effects (acid, sticky).
 - `LAYER_MARKERS` for spawn points and goal markers.

Movement & Collision Logic

- Crawler movement updated to respect **map bounds** and only traverse within painted tiles.
- Movement checks now rely on **tile metadata** (e.g. `walkable`, `digestible`) instead of hardcoding rules.
- Added handling for **walls**:
 - Normal walls block movement.
 - Digestible walls can be passed if the **Acid Sac** upgrade is active.

Hazard System

- Implemented hazard checks when arriving at a tile:
 - **Acid tiles**: apply damage each step, with mitigation via Hardened Skin.
 - **Sticky tiles**: temporarily consume future movement inputs to simulate slowing.

Goal Detection

- Added detection of **Heartroot marker tiles**. Reaching one triggers the win condition.

Code Improvements

- Refactored crawler script with **safe metadata helper functions** (`_get_str`, `_get_int`, `_get_bool`) for robustness.
- Input handling improved to **skip inputs** correctly when slowed.
- Legacy and unused functions were cleaned up or marked for later removal.

In short: Phase 1.3 delivered the **complete tilemap system** with layered tile interactions, crawler movement checks, hazards, and win condition detection. This forms the foundation for further tile-based mechanics in Phase 1.4.